

1. a. $A + B = \begin{bmatrix} 1+2 & 2+1 \\ 3+4 & 4+3 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 7 & 7 \end{bmatrix}$
- b. $2A = \begin{bmatrix} 2 \cdot 1 & 2 \cdot 2 \\ 2 \cdot 3 & 2 \cdot 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$; so $2A - B = \begin{bmatrix} 2-2 & 4-1 \\ 6-4 & 8-3 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 2 & 5 \end{bmatrix}$
- d. $C + D$ is undefined. Since C and D have different dimensions.
- f. $BA = \begin{bmatrix} 2 \cdot 1 + 1 \cdot 3 & 2 \cdot 2 + 1 \cdot 4 \\ 4 \cdot 1 + 3 \cdot 3 & 4 \cdot 2 + 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 13 & 20 \end{bmatrix}$
- g. $AC = \begin{bmatrix} 1 & 4 & 5 \\ 3 & 10 & 11 \end{bmatrix}$
- h. CA is undefined since C has a different number of columns than the rows of A .
2. $AB = \mathbf{0}$ and $BA = \begin{bmatrix} 5 & 5 \\ -5 & 5 \end{bmatrix}$. This means that the vector rows of A are orthogonal to the vector columns of B , but the vector rows of B are not orthogonal to the vector columns of A .
3. Proposition 1.1 is on page 83. Let $A, B, C \in \mathcal{M}_{m \times n}$. We'll denote the ij^{th} entry in a matrix $A + B$ as $(A + B)_{ij}$.
- a. Since real numbers are commutative, the ij^{th} entry of $A + B$ is $(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (A + B)_{ij}$. So the $A + B = B + A$.
- b. Since real numbers are associative, the ij^{th} entry of $((A + B) + C)_{ij} = (a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}) = (A + (B + C))_{ij}$. Thus $(A + B) + C = A + (B + C)$.
- c. $a_{ij} + 0 = a_{ij}$. So then $(A + \mathbf{0})_{ij} = a_{ij} + 0 = 0 + a_{ij} = a_{ij} = A_{ij}$. Thus $A + \mathbf{0} = A$.
6. Both of these statements are false. So we need only provide counterexamples.
- a. Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. We then have $AB = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ and $AC = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. So $AB = AC$ but $A \neq C$ thus the claim is false. Note that this shows that cancellation does not hold for matrix multiplication.
- b. Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. So then $A^2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ but $A \neq I_2$ and $A \neq \mathbf{0}$.